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(22) Date of filing of Application :31/01/2024

(21) Application No.202441006342 A

(43) Publication Date : 16/02/2024

(54) Title of the invention : DEEP LEARNING BASED OVARIAN CANCER DETECTION USING SPECTRAL DOMAIN OPTICAL COHERENCE TOMOGRAPHY

(51) International classification :G06N20/00, G06N3/08, G06T7/00, G06T7/13 (86) International Application No :NA Filing Date :NA (87) International Publication No : NA (61) Patent of Addition to Application Number :NA Filing Date :NA (62) Divisional to Application Number :NA Filing Date :NA	(71)Name of Applicant : 1)RAMCO INSTITUTE OF TECHNOLOGY Address of Applicant :KRISHNAPURAM PANCHAYAT, NORTH VENGANALLUR VILLAGE, RAJAPALAYAM, TAMIL NADU-626117. ----- Name of Applicant : NA Address of Applicant : NA (72)Name of Inventor : 1)Dr.M.GOMATHY NAYAGAM Address of Applicant :RAMCO INSTITUTE OF TECHNOLOGY, ASSOCIATE PROFESSOR & HEAD, CSBS, RAJAPALAYAM, TAMILNADU-626117. ----- 2)Dr.S.ERANA VEERAPPA DINESH Address of Applicant :RAMCO INSTITUTE OF TECHNOLOGY, ASSISTANT PROFESSOR, CSBS, RAJAPALAYAM, TAMILNADU-626117. ----- 3)K.USHARANI Address of Applicant :RAMCO INSTITUTE OF TECHNOLOGY, ASSISTANT PROFESSOR, CSBS, RAJAPALAYAM, TAMILNADU-626117. ----- 4)B.YAZHINI Address of Applicant :RAMCO INSTITUTE OF TECHNOLOGY, ASSISTANT PROFESSOR, CSBS, RAJAPALAYAM, TAMILNADU-626117. ----- 5)M.RUMANA NACHIYAR Address of Applicant :RAMCO INSTITUTE OF TECHNOLOGY, STUDENT, CSBS, RAJAPALAYAM, TAMILNADU-626117. ----- ----- 6)T.SANJANASRI Address of Applicant :RAMCO INSTITUTE OF TECHNOLOGY, STUDENT, CSBS, RAJAPALAYAM, TAMILNADU-626117. ----- ----- 7)V.AKASH Address of Applicant :RAMCO INSTITUTE OF TECHNOLOGY, STUDENT, CSBS, RAJAPALAYAM, TAMILNADU-626117. ----- ----- 8)T.KISHOR Address of Applicant :RAMCO INSTITUTE OF TECHNOLOGY, STUDENT, CSBS, RAJAPALAYAM, TAMILNADU-626117. ----- -----
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(57) Abstract :

Nowadays, ovarian cancer can be detected in the early stages and treated with a better diagnosis improving substantial accuracy and effective treatment. It is capable of detecting ovarian cancer in a nonsurgical way with acceptable specificity and sensitivity in the developing stage. This helps in improving the life of the patient and the impact found in the patient's treatment. This process requires machine learning, image processing, and imaging with high-quality help in accepting the automatic diagnosis. Machine learning applications never helped in analyzing images of SD-OCT based on the knowledge of literature documents. In this work, spectral domain optical coherence tomography has been used to detect ovarian cancer using convolutional neural network (CNN) and bidirectional long short-term memory. Based on the results our diagnosis achieves better performances in achieving no manual tuning or crafting, in the images of Spectral-domain optical coherence tomography (SD-OCT)

No. of Pages : 9 No. of Claims : 4

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"FORM 1" THE PATENTS ACT 1970 (39 of 1970) and THE PATENTS RULES, 2003 APPLICATION FOR GRANT OF PATENT (Sec section 7, 54 and 135 and sub-rule (1) of rule 20)				(FOR OFFICE USE ONLY)	
				Application No.	202441006342
				Filing date:	31-01-2024
 710054770				Amount of Fee paid:	1750/-
				CBR No:	6294
				Signature:	<i>[Signature]</i> 31/01/24
1. APPLICANT'S REFERENCE / IDENTIFICATION NO. (AS ALLOTTED BY OFFICE)					
2. TYPE OF APPLICATION [Please tick () at the appropriate category]					
Ordinary (☒)		Convention ()		PCT-NP ()	
Divisional ()	Patent of Addition ()	Divisional ()	Patent of Addition ()	Divisional ()	Patent of Addition ()
3A. APPLICANT(S)					
Name in Full		Nationality	Country of Residence	Address of the Applicant	
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				City	Rajapalayam
				State	Tamilnadu
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				Pincode	626117
3B. CATEGORY OF APPLICANT [Please tick () at the appropriate category]					
Natural Person (☐)		Other than Natural Person			
		Educational Institution (☒)	Small Entity ()	Startup ()	Others (☐) Educational Institution
4. INVENTOR(S) [Please tick () at the appropriate category]					
Are all the inventor(s) same as the applicant(s) named above?		Yes ()		No (☒)	

If "No", furnish the details of the inventor(s)				
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			Country	India
			Pincode	626117.
5. TITLE OF THE INVENTION:				
“DEEP LEARNING BASED OVARIAN CANCER DETECTION USING SPECTRAL DOMAIN OPTICAL COHERENCE TOMOGRAPHY”				
6. AUTHORISED REGISTERED PATENT-AGENT(S)		IN/PA No.		
		Name		
		Mobile No.		
7. ADDRESS FOR SERVICE OF APPLICANT IN INDIA		Name		Head of the Department, CSBS
		Postal Address		Department of CSBS, Ramco Institute of Technology, Krishnapuram Panchayat, North Venganallur Village,

31-Jan-2024/153555/202441006342/Form 1

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COUNTRY, PARTICULARS OF CONVENTION APPLICATION

COUNTRY, PARTICULARS OF CONVENTION APPLICATION					
Country	Application Number	Filing date	Name of the applicant	Title of the invention	IPC (as classified in the convention country)

9. IN CASE OF PCT NATIONAL PHASE APPLICATION, PARTICULARS OF INTERNATIONAL APPLICATION FILED UNDER PATENT CO-OPERATION TREATY (PCT)

(PCT)	
International application number	International filing date

10. IN CASE OF DIVISIONAL APPLICATION FILED UNDER SECTION 16, PARTICULARS OF ORIGINAL (FIRST) APPLICATION	
1. Name of the Applicant	
2. Name of the Invention	
3. Date of Filing of the Application	
4. Name of the Inventor	
5. Name of the Agent	
6. Name of the Attorney	
7. Name of the Counsel	
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98. Name of the Opponent's Agent	
99. Name of the Opponent's Attorney	
100. Name of the Opponent's Counsel	

ORIGINAL (FIRST) APPLICATION	
Original (first) application No.	Date of filing of original (first) application

11. IN CASE OF PATENT OF ADDITION FILED UNDER SECTION 54, PARTICULARS OF MAIN APPLICATION OR PATENT	
11.1. Name of the applicant	
11.2. Name of the inventor	
11.3. Name of the assignee	
11.4. Name of the licensee	
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11.100. Name of the proprietor	

APPLICATION OR PATENT	
Main application/patent No.	Date of filing of main application

12. DECLARATIONS

Declaration by the inventor(s)

(In case the applicant is an assignee: the inventor(s) may sign herein below or the applicant may upload the assignment or enclose the assignment with this application for patent or send the assignment by post/electronic transmission duly authenticated within the prescribed period).

I/ We, the above named inventor(s) is/are the true & first inventor(s) for this Invention and declare that the applicant(s) herein are our assignee or legal representative.

Date 30.01.2026

Signature(s)

Name(s) Dr. H. Gomathy Nayar Dr. S. Bruna Keerappa Dinesh Ms. K. Usharani Mrs. B. Yazhini

M. R. Nahi'yar.

M.S.M. Rumana Natchiyar

T. Sanjansky

MS. T. Sanjana Sri

~~V. A. A. A.~~

Mr. V. Akash

T. / Gishon

Mr. T. Kishor

(ii) Declaration by the applicant(s) in the convention country

(In case the applicant in India is different than the applicant in the convention country: the applicant in the convention country may sign herein below or applicant in India may upload the assignment from the applicant in the convention country or enclose the said assignment with this application for patent or send the assignment by post/electronic transmission duly authenticated within the prescribed period)

I/We, the applicant(s) in the convention country declare that the applicant(s) herein is/are my/our assignee or legal representative.

Date : 30/01/24.

Signature(s)

Name(s) of the signatory

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Declaration by the applicant(s) 626 117.

We the applicant(s) hereby declare(s) that: -

- o We are in possession of the above-mentioned invention.
- o The provisional/complete specification relating to the invention is filed with this application.
- o The invention as disclosed in the specification uses the biological material from India and the necessary permission from the competent authority shall be submitted by me/us before the grant of patent to me/us.
- o There is no lawful ground of objection(s) to the grant of the Patent to me/us.
- o We are the true & first inventor(s).
- o We are the assignee or legal representative of true & first inventor(s).
- o The application or each of the applications, particulars of which are given in Paragraph-8, was the first application in convention country/countries in respect of my/our invention(s).
- o We claim the priority from the above mentioned application(s) filed in convention country/countries and state that no application for protection in respect of the invention had been made in a convention country before that date by us or by any person from which We derive the title.
- o Our application in India is based on international application under Patent Cooperation Treaty (PCT) as mentioned in Paragraph-9.
- o The application is divided out of my /our application particulars of which is given in Paragraph-10 and pray that this application may be treated as deemed to have been filed on DD/MM/YYYY under section 16 of the Act.
- o The said invention is an improvement in or modification of the invention particulars of which are given in Paragraph-11.

13. FOLLOWING ARE THE ATTACHMENTS WITH THE APPLICATION

(a) Form 2

Item	Details	Fee	Remarks
Complete specification	No. of pages 03	RS 8900/-	

No. of Claim(s)	No. of claims 04 and No. of pages 01		
Abstract	No. of pages 01		
No. of Drawing(s)	No. of drawings 03 and No. of pages 02		

In case of a complete specification, if the applicant desires to adopt the drawings filed with his provisional specification as the drawings or part of the drawings for the complete specification under rule 13(4), the number of such pages filed with the provisional specification are required to be mentioned here.

Complete specification (in conformation with the international application)/as amended before the International Preliminary Examination Authority(IPEA),as applicable(2copies).

Sequence listing in electronic form

Drawings (in conformation with the international application)/as amended before the International Preliminary Examination Authority (IPEA), as applicable (2copies).

Priority document(s) or a request to retrieve the priority document(s) from DAS (Digital Access Service) if the applicant had already requested the office of first filing to make the priority document(s) available to DAS.

Translation of priority document/Specification/International Search Report/International Preliminary Report on Patentability.

Statement and Undertaking on Form 3

Declaration of Inventorship on Form 5

(i) Power of Authority

(j).....

Total fee ₹..890.0/- in Cash/ Banker's Cheque /Bank Draft bearing No ..281599.....Date 30.01.2024...on.....Federal..... Bank.

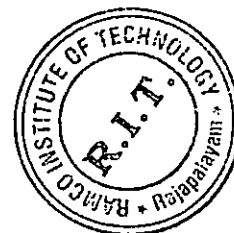
I/We hereby declare that to the best of our knowledge, information and belief the fact and matters stated herein are correct and We request that a patent may be granted to me/us for the said invention.

Dated this Tuesday... Day of 30/01/24.

Signature:

Name:

To, The Controller of Patents
The Patent Office at **PRINCIPAL**
Rajaco Institute of Technology
Note: - Rajapalayam - 626 117.



- * Repeat boxes in case of more than one entry.
- * To be signed by the applicant(s) or by authorized registered patent agent otherwise where mentioned.
- * Tick (□)/cross (x) whichever is applicable/not applicable in declaration in paragraph-12.
- * Name of the inventor and applicant should be given in full, family name in the beginning.
- * Strike out the portion which is/are not applicable.
- * For fee: See First Schedule";

FORM 2 THE PATENT ACT 1970 (39 of 1970) & The Patents Rules, 2003 PROVISIONAL/COMPLETE -SPECIFICATION (See Section 10 and rule 13)	
1.TITLE OF INVENTION: “ DEEP LEARNING BASED OVARIAN CANCER DETECTION USING SPECTRAL DOMAIN OPTICAL COHERENCE TOMOGRAPHY”	
2.APPLICANT(S): a) NAME : Ramco Institute of Technology b) NATIONALITY : Indian c) ADDRESS : Department of Computer Science and Business Systems, Krishnapuram Panchayat, North Venganallur Village, Rajapalayam-626117.	
3.PREAMBLE TO THE DESCRIPTION	
PROVISIONAL Not Applicable	COMPLETE The following specification describes the Invention and the way it performs.
4. DESCRIPTION Description is enclosed.	
5. CLAIMS Claims are enclosed in separate sheet.	
6.DATE AND SIGNATURE Provided at the end of the description.	
7. ABSTRACT OF THE INVENTION Abstract is enclosed in separate sheet.	



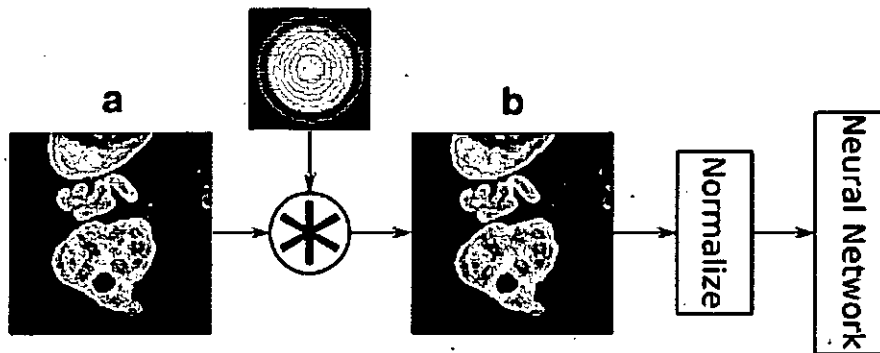


Fig 1: Flowchart for Pre-processing Operation

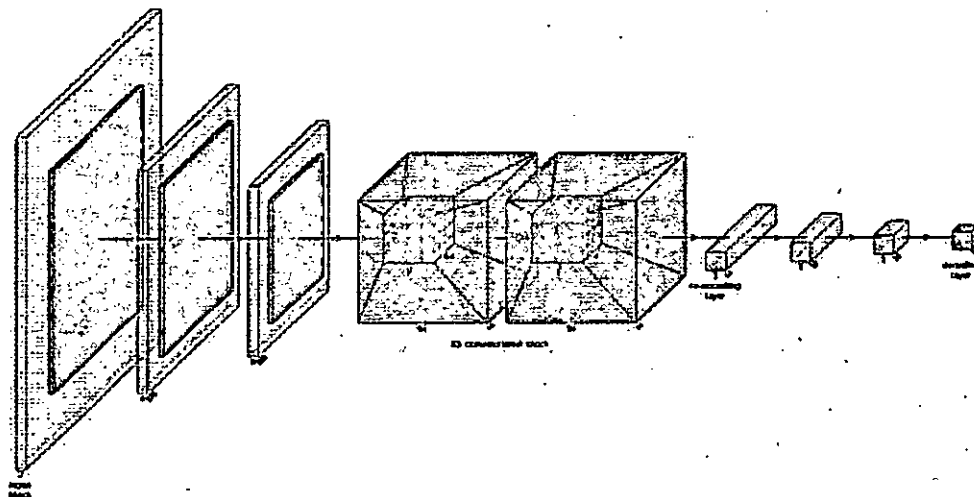


Figure 2: 3D CNN investigation

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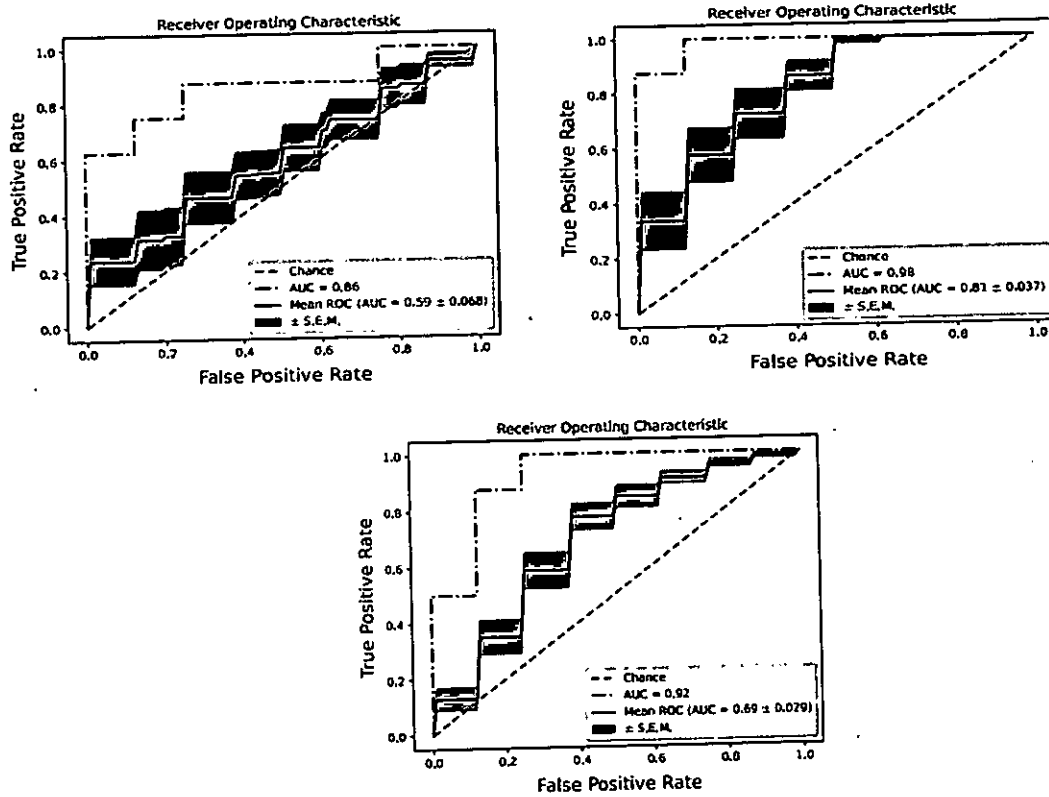


Fig 3: Performance Comparison of model diagnostic efficiency using VGG, Bi-directional LSTM, 3D-CNN

Model	Peak AUC	Mean AUC $\pm$ SE
Area under ROC		
VGG	0.86	0.59 ± 0.068
Bidirectional LSTM	0.98	0.81 ± 0.037
3D CNN	0.92	0.69 ± 0.029

Table 1: standard error for peak AUC and average AUC

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ATENT OFFICE CHENNAI 01/02/2024

Draft Report

1. Title

Deep Learning Based Ovarian Cancer Detection Using Spectral Domain Optical Coherence Tomography.

2. Background of the Invention

a. Field of invention

- One of the most regular diseases that influence women is ovarian disease. Ovarian cancer is the fifth most disease that occurs among women in India. Ovarian cancer is responsible for malignancy regenerative for females for passing great number. Ovarian cancer is present in the ovaries over tumors.
- Based on the nonspecific symptoms of particular devastation in ovarian cancer, idiotically harmless is considered as assessed in isolation. The lacking of early-use; screening tools have a compounded impact on cancer, leading to 80% of late diagnoses.
- According to studies, in many countries, cancer is one of the leading diseases that cause lots of death and people who are affected by cancer was increased over the lifetime percentages.
- The proposed system is used to detect ovarian cancer based on spectral domain optical coherence tomography using convolutional neural network (CNN) and bidirectional long short-term memory.

b. Prior art of state

- From a model transgenic mouse, we collect the data image that developed spontaneously in which a female with ovarian cancer in bilateral epithelial. All developed bilateral ovarian cancer epithelial under female mice under all tag positive, by eight weeks of age in the ovaries evident with an invasive tumor under all mice. Within 8 weeks for imaging, we sacrificed 16 mice and used the system of SD-OCT as were images of explanted organs.
- In previous research, they can be found surgical explanations and breeding protocol on the mouse. We analyzed all images of tissue was immunohistochemistry and for the present by a pathologist as evaluated

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and tumor extent, the TAg protein presence, and cell morphology have been determined. By 8 weeks of age that exhibit ovarian cancer under the TAg mice through the validation under the process provided.

- **Duhit Sengupta et.al (2022)** have demonstrated nuclear morphology under ovarian cancer for accurate diagnosis under a deep hybrid learning pipeline. The malignant potential of cell cancer to analyze by pathologists that adopted an approach for clinical diagnosis under potent determining factor under nuclear feature morphology. In cancer cells the nucleus has been considered under structure alteration, developing techniques of machine learning in a different group based on information from nuclear morphometric like size, nucleus cytoplasm ratio, non-parametric and nuclear shape has been tested under image tissues for different cancer diagnosis. Normal tissues of ovarian cancer exhaustive imaging that have performed ovarian cancer and the matrices of parameter morphometric that can combine a dual pipeline architecture with techniques of deep learning from image processing.
- **Mansi Mathur et.al (2021)** have experimented ovarian cancer for detecting malignancy under the CNN approach. For initial discovery, they required improved techniques and general analysis based on the endurance rate of growth of ovarian malignant. Discovery of disease has to be done in the initial stage to reduce the disease rate. This work productively and execution dependent were clarified under other prior examinations and tentative results.
- **Raina M. Ghoniem et, al (2021)** have demonstrates ovarian cancer diagnosis with multimodal evolution of the DL model. In predicting OC stages better performances proved in deep learning. Employed single modality data under the deep learning model is based on insufficient information about OC characteristics. The model construction to optimize the deep learning to deploy and train that required high-cost computation. To process scene longitudinal data under a predictive antlion optimization of LSTM is included. By the antlion optimization

[Handwritten signature]

algorithm, it automatically set under each customized feature network under topology that make it realize better performances.

3. Objectives of the invention

- To detect ovarian cancer in a nonsurgical way with acceptable specificity and sensitivity in the developing stage.
- To detect ovarian cancer using CNN and bidirectional long short-term memory.
- To perform better in diagnostics using SD-OCT images, when there is no need for human tuning or crafting.

4. Summary of the invention

From a critical step, the work contributes towards the SD-OCT-based ovarian cancer detection and diagnostic system. Representing the adapt abstract and classifier learn were proposed under the tomograms conducive to radiographic signatures detection of ovarian cancer in SD-OCT imagery without selection of manual features. The result represented high classifier discrimination that can be evolved to unseen data that is generalized as expected. Using the depth-resolved SD-OCT recording of ovaries for detecting cancer is the first model in a demonstration to be introduced in the concept, that has been represented in the challenging medium that inferences base of genotype in both wide-field fluorescence and SD-OCT. using a model classifier of generalized linear in detecting SD-OCT-based ovarian cancer represents a result in malignant detection over promising ovarian tissues. However, using full-field SD-OCT in the imaged ovarian tissue biopsies distinct the effort from ours and classification performances on feature hand-crafted development by ovarian SD-OCT data in human analysis.

5. Brief description of the drawing

Figure 1 shows the flowchart for Pre-processing Operation

Figure 2 shows the 3D CNN investigation

Figure 3 describes the Performance Comparison of model diagnostic efficiency using VGG, Bi-directional LSTM, 3D-CNN.

PATENT OFFICE CHENNAI 01/02/2024 14:21

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Table 1 describes the standard error for peak AUC and average AUC

6. Detailed Description of the invention

The proposed approach uses a convolutional neural network (CNN) and bidirectional long short-term memory to identify ovarian cancer through spectral domain optical coherence tomography.

Step 1: Data Acquisition- Initially data of SD-OCT was initially collected from National Library of Medicine and analysis of 3D texture.

Step 2: Data processing- A sequence of pre-processing transformation as we perform to neural network under these rendering useful images.

Step 3: Based on the neuron that is provided with the learning filter under the input signal as the convolution as they computed, the bank of neurons under the class of ANN in CNN with those output states is examined.

Step 4: As a solution, we presented digital classification for the handwritten problem in CNN, and the computer vision near ubiquitous work has been implemented.

Step 5: the lock of primary convolutional contains layer feed-forward that consists of convolutional 2D spatial, which layer of aggregation under a max-pooling under feed.

Step 6: The central layer that makes up a four-channel makes a 3D CNN that is connected under a 64-channel 3D CNN.

Step 7: For each slice t, 3D convolutional lock as output that is connected to a fully connected sequence of layer feed-forward that re-encoded them for decoding layer classification.

Step 8: The outcome demonstrated strong classifier discrimination, which may be extended to previously unknown data and is expectedly generalized.


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CLAIMS

We Claim,

1. Deep Learning Based Ovarian Cancer Detection Using Spectral Domain Optical Coherence Tomography that assists the females from ovarian cancer malignancy regenerative.
2. The proposed system under the tomograms conducive to radiographic signatures detection of ovarian cancer in Spectral-domain optical coherence tomography (SD-OCT) imagery without selection of manual features.
3. Using the depth-resolved SD-OCT recording of ovaries for detecting cancer and a model classifier of generalized linear represents a result in malignant detection over promising ovarian tissues.
4. According to Claim 2, the proposed approach which is resilient to any change in illumination and is trained to detect the ovarian cancer from the dataset with high accuracy.


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Rajapalayam - 626 117.

DEEP LEARNING BASED OVARIAN CANCER DETECTION USING SPECTRAL DOMAIN OPTICAL COHERENCE TOMOGRAPHY

ABSTRACT

Nowadays, ovarian cancer can be detected in the early stages and treated with a better diagnosis improving substantial accuracy and effective treatment. It is capable of detecting ovarian cancer in a nonsurgical way with acceptable specificity and sensitivity in the developing stage. This helps in improving the life of the patient and the impact found in the patient's treatment. This process requires machine learning, image processing, and imaging with high-quality help in accepting the automatic diagnosis. Machine learning applications never helped in analyzing images of SD-OCT based on the knowledge of literature documents. In this work, spectral domain optical coherence tomography has been used to detect ovarian cancer using convolutional neural network (CNN) and bidirectional long short-term memory. Based on the results our diagnosis achieves better performances in achieving no manual tuning or crafting, in the images of Spectral-domain optical coherence tomography (SD-OCT).

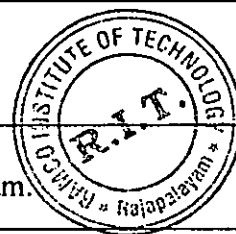

20/1/24
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Rajapalayam - 626 117.

FORM 3
THE PATENTS ACT, 1970
(39 of 1970)
and
THE PATENTS RULES, 2003
STATEMENT AND UNDERTAKING UNDER SECTION 8
(See section 8; Rule 12)



710054772

1. Name of the applicant(s)		I/We a) NAME : Ramco Institute of Technology b) NATIONALITY : Indian c) ADDRESS : Department of CSBS, Krishnapuram Panchayat, North Venganallur Village, Rajapalayam-626117. hereby declare:			
2. Name, Address, Nationality of the joint applicant		(ii) that I/We have not made any application for the same/substantially the same invention outside India. Or (ii) that I/We who have made this application No. dated alone/jointly with made for the same/substantially same invention, application(s) for patent in the other countries, the particulars of which are given below:			
Name of the Country	Date of application	Application No.	Status of the application	Date of publication	Date of grant
3. Name and address of the assignee		(iii) that the rights in the application(s) has/have been assigned to that I/We undertake that up to the date of grant of the patent by the Controller, I/We would keep him informed in writing the details regarding corresponding applications for patents filled outside India within six months from the date of filling of such application. Dated this <u>Tuesday</u> day of <u>30/1/24</u>			
4. To be signed by the applicant or his authorized registered patent agent		Signature x..... PRINCIPAL Dr. L. Ganesan, Principal, Ramco Institute of Technology, Rajapalayam - 626 117.			
5. Name of the natural person who has signed		Dr. L. Ganesan, Principal, Ramco Institute of Technology, Rajapalayam.			
		To The Controller of Patents. The Patent Office, at Chennai.			
Note: Strike out whichever is not applicable.					



FORM 5

THE PATENT ACT 1970

(39 of 1970)

&

The Patents Rules, 2003

DECLARATION AS TO INVENTORSHIP

(See Section 10(6) and rule 13(6))



710054773

1.NAME OF THE APPLICANT(S): Ramco Institute of Technology,
Krishnapuram Panchayat,
North Venganallur Village,
Rajapalayam-626117.

Hereby declare that the true and first inventor(s) of the invention disclosed in the complete specification filed in pursuance of my/our application numbered _____ dated _____ is/are

2.INVENTOR(S):

- a) **NAME** : Dr.M.Gomathy Nayagam
NATIONALITY : Indian
ADDRESS : Department of Computer Science and Business Systems,
Ramco Institute of Technology,
Krishnapuram Panchayat,
North Venganallur Village,
Rajapalayam-626117.
- b) **NAME** : Dr.S.Erana Veerappa Dinesh
NATIONALITY : Indian
ADDRESS : Department of Computer Science and Business Systems,
Ramco Institute of Technology,
Krishnapuram Panchayat,
North Venganallur Village,
Rajapalayam-626117.

c) **NAME** : Ms.K.Usharani

NATIONALITY : Indian

ADDRESS : Department of Computer Science and Business Systems,
Ramco Institute of Technology,
Krishnapuram Panchayat,
North Venganallur Village,
Rajapalayam-626117.

d) **NAME** : Mrs.B.Yazhini

NATIONALITY : Indian

ADDRESS : Department of Computer Science and Business Systems,
Ramco Institute of Technology,
Krishnapuram Panchayat,
North Venganallur Village,
Rajapalayam-626117.

e) **NAME** : Ms.M.Rumana Nachiyar

NATIONALITY : Indian

ADDRESS : Department of Computer Science and Business Systems,
Ramco Institute of Technology,
Krishnapuram Panchayat,
North Venganallur Village,
Rajapalayam-626117.

f) **NAME** : Ms.T.SanjanaSri

NATIONALITY : Indian

ADDRESS : Department of Computer Science and Business Systems,
Ramco Institute of Technology,
Krishnapuram Panchayat,
North Venganallur Village,
Rajapalayam-626117.

g) **NAME** : Mr.V.Akash

NATIONALITY : Indian

ADDRESS : Department of Computer Science and Business Systems,

Ramco Institute of Technology,
Krishnapuram Panchayat,
North Venganallur Village,
Rajapalayam-626117.

h) NAME : Mr.T.Kishor

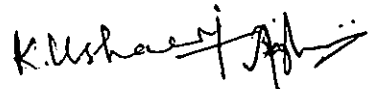
NATIONALITY : Indian

ADDRESS : Department of Computer Science and Business Systems,
Ramco Institute of Technology,
Krishnapuram Panchayat,
North Venganallur Village,
Rajapalayam-626117.

i)

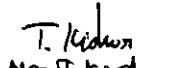
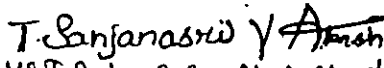
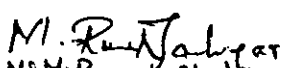
Dated this Tuesday..... Day of 20/01/2024

Signature:



Name of signatory: Dr. M. Gromath Nuyagam Dr. S. Erama Veenappa Ms. K. Ushadani Mrs. B. Yashmi
Dmesh

Signature:



Name of signatory: Ms. M. Ruma Ms. T. Sanjana Sri Mr. V. Akash Mr. T. Kishor

3. DECLARATION TO BE GIVEN WHEN THE APPLICATION IN INDIA IS FILLED BY THE APPLICANT(S) IN THE CONVENTION COUNTRY: -

We the applicant(s) in the convention country hereby declare that our right to apply for a patent in India is by way of assignment from the true and first inventor(s).

Dated this Day of

Signature:

Name of signatory:

4. STATEMENT (to be signed by the additional inventor(s) not mentioned in the application form)

I/We assent to the invitation referred to in the above declaration, being included in the completed specification field in pursuance of the stated application.

Dated this Tuesday Day of 30/01/2024

Signature:

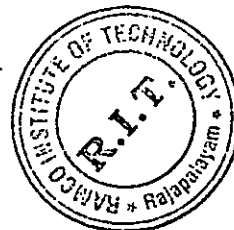
Name of signatory:



PRINCIPAL

Signature:

Ramco Institute of Technology
Rajapalayam - 626 117.



Name of signatory:

To

The Controller of Patent

The Patent Office at Chennai.

Note:

*Repeat boxes in case of more than one entry.

*To be signed by the applicant(s) or by authorized registered patent agent otherwise where mentioned.

*Name of the inventor and applicant should be given in full, family name in the beginning.

*Complete address of the inventor should be given stating the postal indexno./ code, state and country.

*Strike out the column which is/are not applicable.

CBR. NO => 6294

DATE => 31-01-2024

AMOUNT => 2750 x.
31/01/24

[भाग II—खण्ड 3(ii)]

भारत का उपपत्र : असाधारण

91

FORM 9
THE PATENTS ACT,
(39 of 1970)
&
The Patents Rules, 2003
REQUEST FOR PUBLICATION
[See section 11A(2); 24A]



710054774

1. Name, address and nationality of the applicant(s) :

I/We

Name : Ramco Institute of Technology

Nationality : Indian

Address : Department of CSBS,
Krishnapuram Panchayat,
North Venganallur Village,
Rajapalayam-626117.

2. To be signed by the applicant or his authorized registered patent agent

hereby request for early publication of my/our application for patent No.....datedunder section 11A(2) of the Act.

Dated this Tuesday day of 30/01/24

3. Name of the natural person who has signed

Dr.L.Ganesan, Principal,
Ramco Institute of Technology, Rajapalayam.

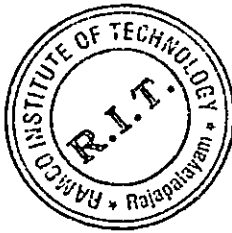
Signature.....

PRINCIPAL

Ramco Institute of Technology
Rajapalayam - 626 117.

To

The Controller of Patents,
The Patent Office,
at Chennai.



Note: - For fee : See First Schedule

PATENT OFFICE CHENNAI 01/02/2024 14:22

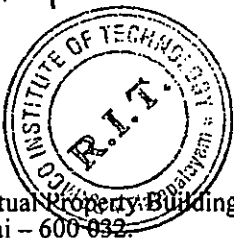
31-Jan-2024/15355/202441006342/Form 9



710054775

[भाग II—खण्ड 3(ii)]

भारत का उपपत्र : असाधारण

FORM 18 THE PATENYS ACT, 1970 (39 of 1970) 8 The Patents Rules, 2003 REQUEST/EXPRES9 REQUEST FOR EXAMINATION OF APPLICATION FOR PATENT [See section 11B and rule 20(4)(ii), 24B(1)(i)]		(FOR OFFICE USE ONLY) RQ. No: <u>820244004440</u> Filing Date: <u>31-01-2024</u> Amount of Fee Paid: <u>44007</u> CBR No: <u>6294</u> Signature: <u>[Signature]</u> <u>31/01/24</u>
1. APPLICANT (S)/OTHER INTERESTED PERSON		
Name	: Ramco Institute of Technology	
Nationality	: Indian	
Address	: Department of CSBS, Krishnapuram Panchayat, North Venganallur Village, Rajapalayam-626117.	
2. Statement In case of request for examination made by the applicant(s)		
I/We hereby request that my/our application for patent no _____ filed on _____ for the invention titled "Deep Learning Based Ovarian Cancer Detection Using Spectral Domain Optical Coherence Tomography" shall be examined under sections 12 and 13 of the Act. (Or) I/We hereby make an express request that my/our application for patent no _____ filed on _____ based on Patent Cooperation Treaty (PCT) application no. _____ dated _____ made in country _____ shall be examined under sections 12 and 13 of the Act, immediately without waiting for the expiry of 31 months as specified in rule 20(4)(ii).		
Statement in case of request for examination made by any other interested person		
I/We the interested person request for the examination of the application no. _____ dated _____ filed by the applicant _____ titled _____ under sections 12 and 13 of the Act. As an evidence of my/our interest in the application for patent following documents are submitted		
4. ADDRESS FOR SERVICE:		
Dated this <u>Tuesday</u> of <u>30/01/24</u>		
To		Signature <u>[Signature]</u> Name of the Signatory PRINCIPAL Ramco Institute of Technology Rajapalayam - 626 117.
The Controller of Patent The Patent Office, at Intellectual Property Building, G.S.T.Road, Guindy, Chennai - 600 032.		
Note: • To be signed by the applicant(s) or by his authorized registered patent agent • Strike out the column which is/are not applicable.		

31-Jan-2024/153555/202441006342/Form 18

PATENT OFFICE CHENNAI 01/02/2024 14:22

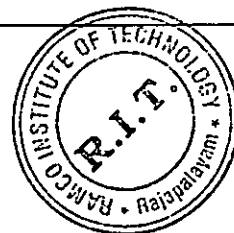


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FORM 28
THE PATENTS ACT, 1970
(39 of 1970)
&
The Patents Rules, 2003
TO BE SUBMITTED BY A SMALL ENTITY /
STARTUP
[See rules 2 (fa), 2(fb) and 7]

1. Insert Name, address and nationality	I/We Ramco Institute of Technology Applicant/patentee in respect of the patent application no. or patent no Hereby declare that I/we am/are a small entity in accordance with rule 2(fa) or a startup in accordance with rule 2(fb) and submit the following document(s) as proof.
2. Documents to be submitted	
i. For claiming the status of a small entity:	
A. For an Indian applicant: Evidence of registration under the Micro, Small and Medium Enterprises Act, 2006 (27 of 2006).	
B. In Case of a foreign entity: Any other document.	
ii. For Claiming the status of a startup.	
iii. For claiming the status of an educational institution	
A. For an Indian applicant: Any document as evidence of eligibility, as defined in rule 2(ca). (All India Council for Technical Education Affiliation Proof)	
B. In Case of a foreign entity: Any other document.	
3. To be signed by the applicant or his authorized registered patent agent	The information provided herein is correct to the best of my/our knowledge and belief. Dated this <u>Tuesday</u> day of <u>30</u> / <u>01</u> / <u>24</u>
4. Name of the natural person who has signed	Signature (Name) Dr. L. Ganesan (Designation) Principal Ramco Institute of Technology Rajapalayam - 625 877. To The Controller of Patents. The Patent Office, at Chennai.

Note: - For fee : See First Schedule



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31-Jan-2024/15355/202441006342/FORM28

All India Council for Technical Education

(A Statutory body under Ministry of Education, Govt. of India)

Nelson Mandela Marg, Vasant Kunj, New Delhi-110070 Website: www.aicte-india.org



APPROVAL PROCESS 2023-24

Extension of Approval (EOA)

F.No. Southern/1-36263839846/2023/EOA

Date: 15-May-2023

To,

The Principal Secretary
(Higher Education) Govt. of Tamil Nadu,
N. K. M. Bld. 6th Floor Secretariat,
Chennai-600009

Sub: Extension of Approval for the Academic Year 2023-24

Ref: Online application of the Institution submitted for Extension of Approval for the Academic Year 2023-24

Sir/Madam,

In terms of the provisions under the All India Council for Technical Education (Grant of Approvals for Technical Education) Regulations, 2020 notified on 4th February 2020 and amended on 24th February 2021 and norms standards, procedures and conditions prescribed by the Council from time to time, I am directed to convey the approval to:

Permanent Id	1-1423258331	Application Id	1-36263839846
Name of the Institution	RAMCO INSTITUTE OF TECHNOLOGY	Name of the Society/Trust	RAJA CHARITY TRUST
Institution Address	NORTH VENGANALLUR VILLAGE, KRISHNAPURAM PANCHAYAT, RAJAPALAYAM, RAJAPALAYAM, VIRUDHUNAGAR, Tamil Nadu, 626117	Society/Trust Address	RAJAPALAYAM, VIRUDHUNAGAR, Tamil Nadu, 626117
Institution Type	Private-Self Financing	Region	Southern
Year of Establishment	2013		

To conduct following Courses with the Intake indicated below for the Academic Year 2023-24

Level	Program	Course	Affiliating Body (University /Body)	Intake Approved for 2022-23	Intake Approved for 2023-24	NRI Approval Status	FN / Gulf quota/ OCI/ Approval Status
UNDER GRADUATE	ENGINEERING AND TECHNOLOGY	ARTIFICIAL INTELLIGENCE AND DATA SCIENCE	Anna University, Chennai	120	120	No	No
UNDER GRADUATE	ENGINEERING AND TECHNOLOGY	CIVIL ENGINEERING	Anna University, Chennai	60	60	No	No

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Page 1 of 4

Application No: 1-36263839846

Note: This is a Computer generated Report. No signature is required.

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Letter Printed On: 25 May 2023

31-Jan-2024/153555/202441006342/FORM28

Level	Program	Course	Affiliating Body (University /Body)	Intake Approved for 2022-23	Intake Approved for 2023-24	NRI Approval Status	FN / Gulf quota/ OCI/ Approval Status
UNDER GRADUATE	ENGINEERING AND TECHNOLOGY	COMPUTER SCIENCE & ENGINEERING	Anna University, Chennai	120	120	No	No
UNDER GRADUATE	ENGINEERING AND TECHNOLOGY	COMPUTER SCIENCE AND BUSINESS SYSTEMS	Anna University, Chennai	60	60	No	No
UNDER GRADUATE	ENGINEERING AND TECHNOLOGY	ELECTRICAL AND ELECTRONICS ENGINEERING	Anna University, Chennai	60	60	No	No
UNDER GRADUATE	ENGINEERING AND TECHNOLOGY	ELECTRONICS AND COMMUNICATIONS ENGINEERING	Anna University, Chennai	120	120	No	No
UNDER GRADUATE	ENGINEERING AND TECHNOLOGY	INFORMATION TECHNOLOGY	Anna University, Chennai	60	60	No	No
UNDER GRADUATE	ENGINEERING AND TECHNOLOGY	MECHANICAL ENGINEERING	Anna University, Chennai	60	60	No	No

It is mandatory to comply with all the essential requirements as given in APH 2023-24 (Appendix 6)


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ALL INDIA COUNCIL FOR TECHNICAL EDUCATION

Application No: 1-36263839846

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01/02/2024 14:22

Important Instructions

1. The State Government/ UT/ Directorate of Technical Education/ Directorate of Medical Education shall ensure that 10% of reservation for Economically Weaker Section (EWS) as per the reservation policy for admission, operational from the Academic year 2019-20 is implemented without affecting the reservation percentages of SC/ ST/ OBC(NCL) / General. However, this would not be applicable in the case of Minority Institutions referred to the Clause (1) of Article 30 of Constitution of India. Such Institution shall be permitted to increase in annual permitted strength over a maximum period of two years.
2. The Institution offering courses earlier in the Regular Shift, First Shift, Second Shift/Part Time are now amalgamated as total intake and shall have to fulfil all facilities such as Infrastructure, Faculty and other requirements as per the norms specified in the Approval Process Handbook 2023-24 for the Total Approved Intake. Further, the Institutions Deemed to be Universities/ Institutions having Accreditation/ Autonomy status shall have to maintain the Faculty: Student ratio as specified in the Approval Process Handbook.
3. Strict compliance of Anti-Ragging Regulation, Establishment of Committee for SC/ ST, Establishment of Internal Committee (IC), Establishment of Online Grievance Redressal Mechanism, Barrier Free Built Environment for disabled and elderly persons, Fire and Safety Certificate should be maintained as per the provisions made in Approval Process Handbook and AICTE Regulation notified from time to time.
4. In case of any differences in content in this Computer generated Extension of Approval Letter, the content/information as approved by the Executive Council / General Council as available on the record of AICTE shall be final and binding.
5. As per the AICTE Notification dated 29.01.2014 and amended thereto, it shall be mandatory for each Technical Education Institution, University Department and Institution Deemed to be University imparting Technical Education to get accreditation (NBA) for at least 60% of the eligible courses in the next ONE (1) Years' time, otherwise EoA for the subsequent Academic Year (A.Y. 2024-25) shall not be issued by the Council
6. Deemed to be University: Institutions Deemed to be Universities (Running Technical Education Programmes), it is mandatory to have AICTE approval from the Academic Year 2018-19 in compliance of the Hon'ble Supreme Court Order dated 03-11-2017 passed in CA No.17869- 17870 /2017.

Prof.Rajive Kumar
Member Secretary, AICTE

Copy to:

1. The Director Of Technical Education**, Tamil Nadu
2. The Registrar**,
Anna University, Chennai
3. The Principal / Director,
RAMCO INSTITUTE OF TECHNOLOGY
North Venganallur Village,
Krishnapuram Panchayat,
Rajapalayam,
Rajapalayam,Virudhunagar,
Tamil Nadu,626117
4. The Secretary / Chairman,

RAJAPALAYAM,VIRUDHUNAGAR
Tamil Nadu,626117
5. Guard File(AICTE)


PRINCIPAL
Ramco Institute of Technology
Rajapalayam - 626 117.

Note: Validity of the Course details may be verified at <http://www.aicte-india.org/>

** Individual Approval letter copy will not be communicated through Post/Email However, a consolidated list of Approved Institutions(bulk) may be downloaded from the respective login id's.

This is a computer generated Statement. No signature Required

31-Jan-2024/153555/202441006342/FORM28

Application No: 1-36263839646

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ALL INDIA COUNCIL FOR TECHNICAL EDUCATION

ATENT OFFICE CHENNAI 01/02/2024 14:22

Page 4 of 4

Letter Printed On: 25 May 2023